IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-253

1. Topic (12 words max)

AI and AR-Powered Assistant for Physical, Mental, Educational and Fetal Health during Pregnancy

2. Research group the project belongs to

MR - Mixed Reality

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Pregnancy is a dynamic physiological and psychological journey that demands holistic monitoring to safeguard both maternal and fetal wellbeing. Pregnancy causes many changes in a woman's body, especially physical health factors such as heart and blood sugar levels. Monitoring blood pressure, blood glucose, and heart rate is very important to detect early signs of complications. High blood pressure during pregnancy can lead to serious problems like preeclampsia, which can harm both mother and baby [1]. Similarly, high blood sugar levels can lead to gestational diabetes, increasing the risk of having a birth complications. Tracking these vital signs regularly can help doctors take early action to avoid such risks. The World Health Organization (WHO) recommends that home monitoring of blood pressure and glucose during pregnancy can improve health outcomes for mothers and babies [3].

Mental health during pregnancy is equally critical. Elevated maternal stress, anxiety, and depression correlate with impaired fetal neurodevelopment and increased risk of preterm birth or low birth weight [2]. Despite guidelines promoting prenatal mental health screening [4], gaps remain in effective, scalable solutions—particularly tools that can passively assess emotional states and proactively predict risk.

For fetal monitoring using cardiotocography and ultrasound is typically performed during occasional clinic visits and relies heavily on the doctor's judgment. In these critical situations, physicians welcome decision support systems that confirm and strengthen their clinical decisions [5]. Advances in AI now allow continuous, real-time fetal assessment in low-resource settings, significantly reducing stillbirths and neonatal deaths [6]. Review studies demonstrate AI's superior performance in electronic fetal monitoring, offering noise suppression, feature extraction, and automated classification compared to classic methods [7]. In addition, emerging edge-AI solutions have enabled smartphone-based fetal assessment during home visits in underserved communities [8].

Yet despite these advances, no integrated solution currently supports comprehensive prenatal care across physical, mental, educational, and clinical fetal health domains—especially leveraging AI with interactive AR to empower women and bridge resource gaps. Existing digital decision support tools increase knowledge and data recording , but fall short of combining immersive education and predictive analytics aligned with local clinical needs.

Pregnant mothers often miss clear, engaging guidance during pregnancy, leading to low health literacy and weaker maternal–fetal bonding, especially in remote or busy settings. By introducing a personalized AR module featuring interactive 3D visualizations, this project addresses those gaps. AR-based prenatal education has been shown to significantly improve mothers' knowledge and health behavior, while immersive 3D fetal imagery boosts maternal–fetal attachment and reduces anxiety [9].

This project addresses these problems by developing an AI and AR-integrated intelligent assistant for pregnancy, combining physical wellness monitoring, emotion-based risk prediction, interactive AR education, and fetal health decision support. While three components directly assist pregnant mothers in daily care, one component is designed to support obstetricians and gynecologists to make decisions on fetal health assessments.

References

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[3] P. T. R. D. K. K. C. E. Z. C. A. L. B. T. Ö. G. P. d. L. R. &. N. M. Yeh, " Self-monitoring of blood pressure among women with hypertensive disorders of pregnancy: a systematic review," *BMC Pregnancy and Childbirth,* vol. 1, no. 1, p. 1, 2022, DOI: 10.1186/s12884-022-04751-7

[4] "Implementing Perinatal Mental Health Screening," acog, [Online]. Available: https://www.acog.org/programs/perinatal-mental-health/implementing-perinatal-mental-health-screening.

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[6] C. Kimeu, "How AI monitoring is cutting stillbirths and neonatal deaths in a clinic in Malawi," theguardian, 2025. [Online]. Available: https://www.theguardian.com/global-development/2024/dec/06/how-ai-monitoring-is-cutting-stillbirths-and-neonatal-deaths-in-a-clinic-in-malawi.

[7] K. M. R. V. K. R. e. a. Barnova, "Artificial Intelligence and Machine Learning in Electronic Fetal Monitoring," *Archives of Computational Methods in Engineering,* vol. 31, p. 2557–2588, 2024, DOI:10.1007/s11831-023-10055-6

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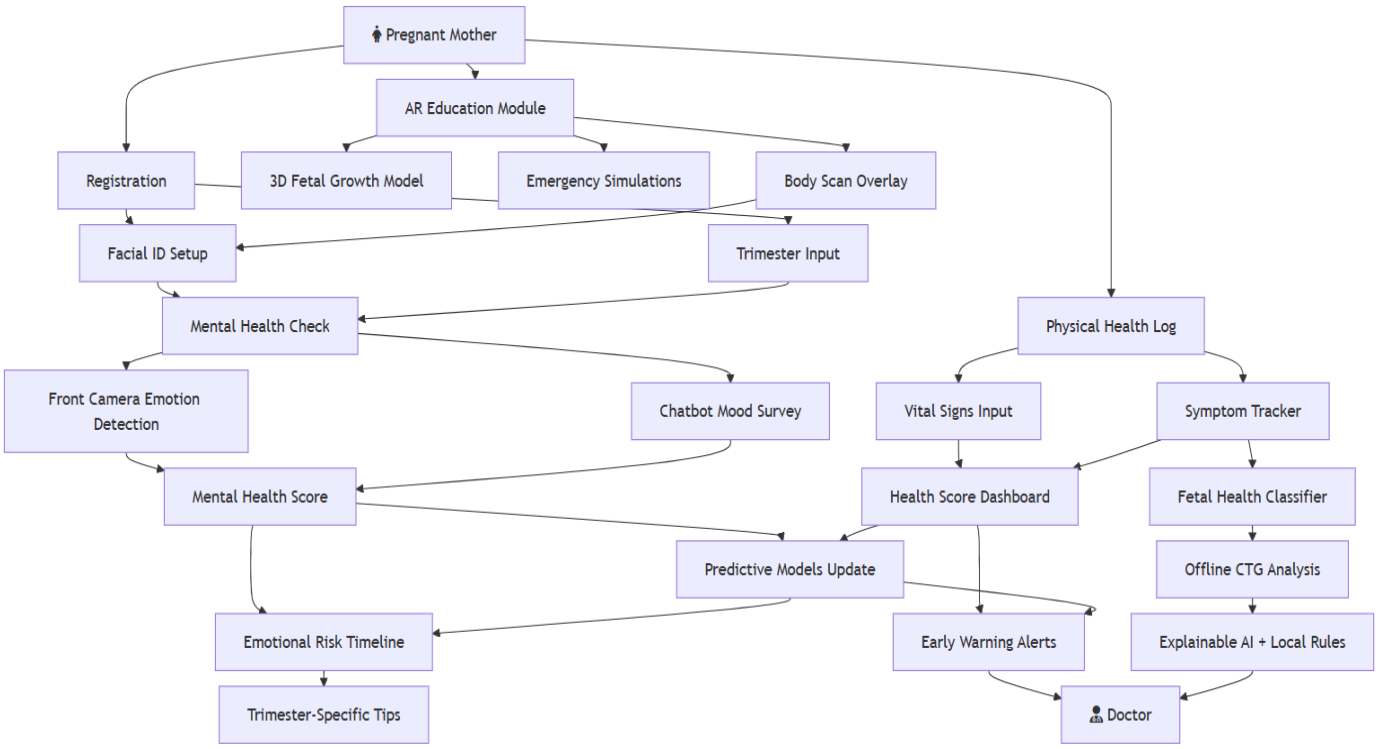
[9] K. N. K. H. J. C. K. C. A. K. B. S. J. M. W. P. J. Y. &. O. K. J. Lee, "Effects of fetal images produced in virtual reality on maternal-fetal attachment: Randomized controlled trial," *Journal of Medical Internet Research,* vol. 25, p. e43634, 2023, DOI: 10.2196/43634

6. Brief description of the nature of the solution, including a conceptual diagram (250 words max)



This research presents a holistic mobile-based solution designed to support pregnant mothers and clinicians through four intelligent and interconnected components. The solution aims to improve maternal–fetal health outcomes by combining physical wellness tracking, Mental Health Monitoring and Risk Prediction, immersive education, and clinician decision support into a single unified system.

The first component enables the monitoring of key physical health parameters such as blood pressure, heart rate, and blood glucose. These inputs are collected manually or via wearable devices, then analyzed to detect abnormal patterns that may pose risk during pregnancy. The second component also uses user's physical health parameters as well as the emotion recognition to assess the psychological well-being of pregnant mothers through text or image inputs, allowing early detection of emotional distress that could impact fetal development.

The third component introduces an immersive AR-based education and emergency preparedness module. It delivers personalized 3D visualizations of fetal development, educational content, and emergency guidance, thereby enhancing health literacy, maternal–fetal bonding, and readiness for critical situations. This component is fully embedded within the mobile interface to ensure interactivity and accessibility.

The fourth and final component is a confidence-aware, explainable decision support system (DSS) for fetal health classification, primarily intended for obstetricians and gynecologists. This component runs as a desktop application that integrates with the mobile app, functioning entirely offline and applying local clinical guidelines to improve decision accuracy in rural or resource-limited settings.

Together, these components create a context-aware, AI-driven ecosystem that enhances both self-care and clinical care throughout the pregnancy journey.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Physical Wellness Monitoring component requires expertise in maternal health parameters such as blood pressure, blood glucose, and heart rate monitoring. Knowledge of threshold values and their clinical significance during pregnancy is essential. IT knowledge is required to design systems for data acquisition from wearable devices or manual input, and to ensure accurate tracking, visualization, and alert generation.

Developing an AI-based Mental Health Monitoring and Risk Prediction system for pregnant women requires basic psychological knowledge combined with skills in computer vision and machine learning. It involves facial emotion recognition using camera input and chatbot-based mood check-ins to assess emotional states like stress and sadness. The system uses labeled facial datasets such as FER-2013 or AffectNet and applies trimester-specific logic to predict emotional risks. It also integrates culturally appropriate wellness suggestions and is optimized for offline use, ensuring accessibility in low-resource settings.

Immersive AR Education and Emergency Preparedness relies on specialized skills in augmented reality, 3D modeling, and contextual content delivery. Domain knowledge in prenatal care education, maternal–fetal bonding, and emergency scenarios is essential to ensure the content is clinically accurate and emotionally engaging. Data requirements include AR assets (e.g., fetal development models) and educational material aligned with local maternal care practices.

Fetal Health Decision Support System component demands knowledge of cardiotocography (CTG) interpretation, fetal health classification, and Sri Lankan clinical guidelines. Expertise in machine learning is required to train classification models, integrate explainable AI (e.g., SHAP), and apply confidence scoring. The system also needs offline deployment and local rule integration, requiring understanding of software packaging and clinical decision logic. A dataset containing features extracted from Cardiotocogram (CTG) examinations, along with expert-classified labels, is required to meet the data requirements.

8. Objectives and Novelty

Main Objective
This research aims to develop an AI-driven mobile and desktop-integrated pregnancy care system that supports both pregnant mothers and clinicians by providing mental health risk prediction, physical wellness monitoring, immersive AR-based education and emergency preparedness, and a decision support system for fetal health classification.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Perera A.N.M IT21215438	AI-Based Mental Health Monitoring and Risk Prediction for Pregnant Women	• Develop a facial expression recognition system that analyzes user-uploaded images to detect emotional states such as stress, anxiety, and sadness. • Design and implement a chatbot to ask simple, pregnancy-related mental health questions and collect emotional feedback from the user. • Create a fusion algorithm that combines chatbot responses and facial expression analysis to calculate a personalized mental health score. • Build a rule-based recommendation engine to provide personalized wellness suggestions such as guided meditation, breathing exercises, mood-boosting activities, and therapeutic games based on the user's detected emotional state and mental health score. • Design a user-friendly mobile interface that guides	This component introduces a unique mental health monitoring approach specifically for pregnant women by combining chatbot-based emotional screening with facial expression analysis. Unlike existing systems that rely on a single method, this hybrid model improves accuracy by analyzing both internal feelings (via chatbot questions) and external expressions (via a selfie taken through the phone's camera). The system generates a personalized mental health score based on the combined results, helping identify emotional risks such as stress, anxiety, or sadness. Based on the score, it provides personalized mental wellness support like calming exercises, short

		the user through the chatbot, image upload, result display, and personalized suggestions.	meditations, or light cognitive games to stabilize the user's emotional state. This real-time, dual-input emotion detection and personalized response system is a novel contribution, as current solutions do not offer this level of tailored emotional care specifically for pregnant women.
Jayasinghe N. IT22924650	To develop an immersive and context-aware AR module that delivers personalized pregnancy education and emergency preparedness support for expectant mothers, enhancing understanding, maternal-fetal bonding, and health literacy through interactive 3D visualization.	• **Generalized 3D Fetal Growth Models** Displays trimester-specific 3D fetal models on detected flat surfaces using AR. Users can rotate and zoom in to explore fetal size and posture at different stages of development. • **Emergency Preparedness AR Simulations** Simulates critical labor-related emergencies such as water breaking, vaginal bleeding, and contractions through branching AR scenes that include visual guidance and step-by-step instructions. • **Voice-Narrated Visual Guides** Adds Sinhala, Tamil, and English voice narration to all AR scenes, guiding users through each experience regardless of reading ability. • **Personalized Body Scan with Body Metrics-Based Scaling (ML-powered)** Uses the camera to extract body dimensions in real time and scales a	This AR module combines real-time anatomical overlays with personalized body metrics scaling for pregnancy education. It offers trimester-based 3D visualizations and interactive emergency simulations tailored to the user's body. It uniquely integrates AI-powered body dimension detection with immersive AR in maternal health and provides multilingual voice guidance for low-literacy users. Designed for accessibility in low-resource settings like Sri Lanka, it addresses a critical gap in personalized, visual, and context-aware maternal education.

		generalized 3D internal organ model accordingly. Trimester-based changes are shown through a timeline slider in the 3D model.	
B.P Buddika IT22312808 Explainable Decision Support System for Fetal Health Classification with Offline Capability	A Confidence-Aware, Explainable Decision Support System for Fetal Health Classification with Offline Capability and Local Guideline Adaptation.	• System Planning Outline the core objectives and features of the proposed decision support system based on previously identified research gaps. Finalize CTG(Cardiotocograms) inputs, expected outputs, and integration points for explainability, confidence scoring, local rule adaptation, and offline use. • Model Development Use the fetal health classification datasets to train models such as Random Forest and XGBoost. Develop an ensemble model to improve prediction reliability. Add confidence scoring to enhance clinical trust and highlight uncertain predictions. • Explainable AI Integration Integrate SHAP to provide explanations for each prediction. Highlight key variables influencing the classification to improve clinical transparency. Present results visually and in text to support decision-making and build trust among healthcare users. • Local Medical Guideline Adaptation Consult with Sri Lankan clinicians to understand local thresholds and response practices. Implement a rule-based layer on top of ML predictions to adapt outputs to local healthcare needs. • Offline Capability	This research component introduces a context-aware decision support system that adapts a globally trained fetal health classification model to the Sri Lankan clinical environment by integrating local medical guidelines into its decision-making logic. In addition, the system is designed to run fully offline with an offline mode, ensuring accessibility in rural and low-resource healthcare settings where internet connectivity and advanced infrastructure are limited. Together, these features enable the solution to provide regionally relevant, explainable, and reliable fetal health assessments for frontline consultants, even in remote areas.

		Build an offline-capable application that accepts CTG derived variables and outputs fetal health classification. Ensure usability in rural settings with limited or no internet access. • Evaluation Evaluate model performance using accuracy, F1-score, and interpretability. Validate SHAP-based explanations with sample cases. Test local rule integration for correctness and usefulness in clinical workflows.	
Wijethunga D.N IT22306722	Physical Health Monitoring and Wellness guidance for Pregnant Mothers	• Vital Signs Monitoring- allows pregnant mothers to record their critical health parameters, which are important for detecting health risks during pregnancy. • Early Warning System- Detects early signs of complications like preeclampsia, gestational diabetes, abnormal weight gain, and preterm labor. Sends real-time alerts with explanations to mothers and doctors via app or SMS. • Personalized Health Insights- Gives weekly health tips based on the mother's condition, with trimester-based focus on rest, nutrition, activity, and stress management. • Visual Health Score Dashboard- Displays weekly health trends with color-coded status, key factor explanations, and an	This component introduces a predictive physical health monitoring system that leverages machine learning models trained on maternal health data to assess the overall wellness of pregnant mothers. By integrating multimodal inputs, including vital signs, daily symptom logs, activity and sleep data, nutritional intake, and (fetal movement reports), the system delivers a personalized physical health status categorized as **Healthy**, **Monitor Closely**, or **High Risk**. Utilizing classification models such as Random Forest or XGBoost, alongside time-series models like LSTM or GRU, the system continuously analyzes health trends while employing SHAP for interpretable and transparent predictions. Designed for adaptability

		exportable summary for doctors.	and personalization, this solution empowers both mothers and healthcare providers with timely, explainable insights, enabling early intervention and informed prenatal care decision-making.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Perera A.N.M IT21215438	My mental health monitoring component aligns with the Information Technology specialization by applying key areas such as artificial intelligence, mobile computing, human-computer interaction, and offline-capable system design. The component leverages AI through facial expression recognition models to detect emotional states, integrates lightweight mobile-friendly architecture to support offline environments, and follows software engineering principles to create a modular, scalable mental health support system. It also reflects IT competencies in user-centered design, emotion data processing, predictive logic development, and the adaptation of global models to local cultural contexts, showcasing the practical use of technology to address real-world healthcare challenges.
Jayasinghe N. IT22924650	My individual component is the Augmented Reality (AR) Module, applying core IT principles to tackle maternal healthcare challenges. It uses mobile AR to deliver immersive pregnancy education through fetal development visualizations, maternal body transformations, and emergency scenario simulations. Development requires skills in Unity with ARCore/ARKit, 3D modeling, and animation integration key to mobile and interactive system design. The module includes AI-based body metrics estimation using live camera input to personalize 3D anatomical overlays, demonstrating machine learning and computer vision applications within IT. It ensures privacy and real-time interactivity, highlighting IT's role in building safe, adaptive digital experiences. Additionally, the AR module promotes accessibility with multilingual voice narration (Sinhala, Tamil, English), overcoming language and literacy barriers. It incorporates Human-Computer Interaction (HCI) and usability principles for intuitive engagement with health content. Overall, this component integrates software engineering, AI, AR, and user-centered design into a powerful healthcare education tool.
B.P Buddika IT22312808	My fetal health classification research component aligns closely with the Information Technology specialization by applying core concepts such as machine learning, decision support systems, software engineering, and offline-first application development. It demonstrates the practical integration of AI models into real-world systems, the use of Explainable AI for transparency, and the development of a standalone application that functions without internet connectivity. The project also reflects IT skills in data

	preprocessing, model deployment, system design, and context-aware application logic
Wijethunga D.N IT22306722	My physical health monitoring component introduces a machine learning–based system designed to predict the overall physical health status of pregnant mothers. It analyzes a combination of data sources, including vital signs such as heart rate, blood pressure, and weight, daily symptom logs, activity and sleep data from wearables, nutritional intake, and (fetal movement). The system includes a Vital Signs Monitoring feature, allowing pregnant mothers to record their critical health parameters, which are essential for identifying early health risks during pregnancy. Based on all collected inputs, the system classifies the mother's condition as Healthy, Monitor Closely, or High-Risk using models like Random Forest, XGBoost, or LSTM, with SHAP used to explain predictions in a transparent manner. An integrated Early Warning System detects potential complications such as preeclampsia, gestational diabetes, or risk of preterm labor and sends real-time alerts via app or SMS to both mothers and healthcare professionals. The system also provides Personalized Health Insights, including trimester-specific wellness tips and lifestyle recommendations, and presents a Visual Health Dashboard that displays weekly health scores with clear, color-coded trends. The novelty of this component lies in its ability to deliver predictive, explainable, and personalized maternal care using easily accessible, non-invasive data—without the need for advanced medical equipment.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Mahima	Weerasinghe	
Co-Supervisor	Ms.	Osuri	Dunuwila	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Novelties have to be further refined beyond application novelty

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	✓
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Provide the novelties with inline citations.
Use quantitative figures to build scientific arguments.

Staff Member's Name	Signature

***Important**:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.